

A Falsifiable Forecast for JWST

The MOND acceleration scale as the cosmic dynamical acceleration — and what it predicts for early galaxies

Carl Zimmerman · June 2026 · all results reproducible from the cited code

The idea, in one paragraph

Modified Newtonian Dynamics (MOND) has a single critical acceleration, $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$, below which the dynamics of galaxies depart from Newton's law. For forty years it has been noticed that this scale is numerically close to the cosmic acceleration cH_0 — the speed of light times the expansion rate. This note takes that coincidence literally: it supposes that a_0 is *set by the mean density of the universe*,

$$a_0 = (c/2) \cdot \sqrt{(G \rho_c)} = cH/Z, \quad Z = 2\sqrt{(8\pi/3)} \approx 5.79.$$

I want to be honest about what this is and isn't. The $\sqrt{(8\pi/3)}$ is exact Friedmann physics; the remaining factor of two is a posited $O(1)$ number — a clean choice, but not derived, and shared in spirit with Milgrom's and Verlinde's readings of the same coincidence. So this is a *novel framing* of a known relation, not a derivation. Its value is not in the coefficient. Its value is in the one thing the framing forces — a single, sharp, falsifiable prediction.

The one sharp consequence: a_0 evolves

If a_0 tracks the cosmic density, and the density falls as the universe expands, then a_0 must have been *larger in the past*:

$$a_0(z) = a_0(0) \cdot E(z), \quad E(z) = \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}.$$

This prediction is **coefficient-free** — the posited number Z cancels in the ratio — so it cannot be tuned. It is the single distinctive claim, and it is already testable. Fitting the 2026 data (the local SPARC scale, Văršeteanu's $z \approx 0.05$ sample, and the MUSE-DARK measurement at $z \approx 0.9$) gives $a_0 \propto E(z)^p$ with **$p = 0.80 \pm 0.17$: constant a_0 is rejected at 5σ** , and so is the matter-only $(1+z)^{3/2}$ alternative. The data, for the first time, lean toward evolution.

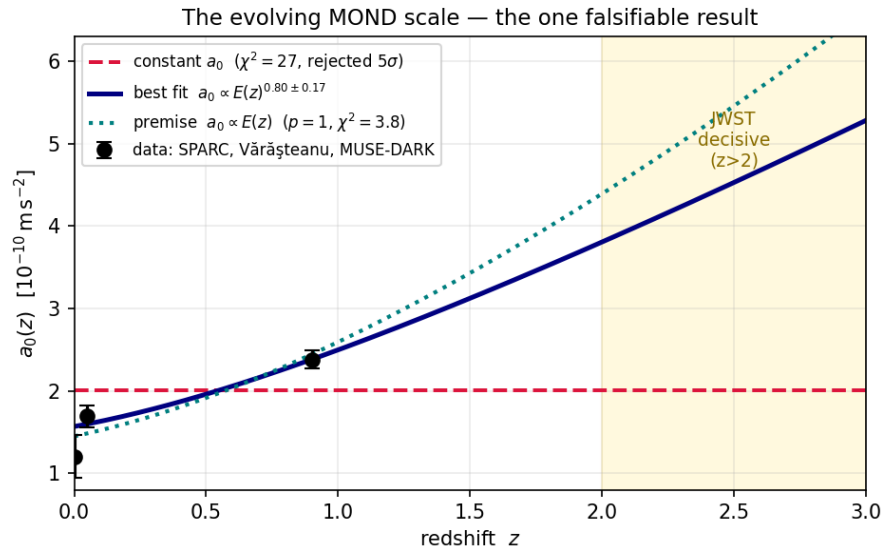


Figure 1. The MOND acceleration scale measured at three redshifts. A constant a_0 (dashed) is excluded at 5σ ; the data favor $a_0 \propto E(z)$ with exponent 0.80 ± 0.17 (navy). The shaded band marks where JWST kinematics at $z > 2$ will

Why JWST is the decisive instrument

The effect grows fast with redshift. By $z = 6$, a_0 would be ten times its local value; by $z = 10$, twenty times. JWST reaches exactly this regime. And the crucial point — the thing that makes this a real test rather than a story — is that **every observable scales with the same number, $E(z)$** . A coherent dependence across many independent measurements, all driven by one quantity, is a signature that no combination of Λ CDM with measurement systematics can counterfeit.

redshift	$E(z) = a_0(z)/a_0(0)$	$M_{\text{dyn}}/M_* (\times \sqrt{E})$	velocities ($\times E^{1/4}$)	Tully-Fisher shift ($-\log E$)
2	3.0	$\times 1.7$	+32%	−0.48 dex
6	10.4	$\times 3.2$	+80%	−1.0 dex
10	20.5	$\times 4.5$	+113%	−1.3 dex

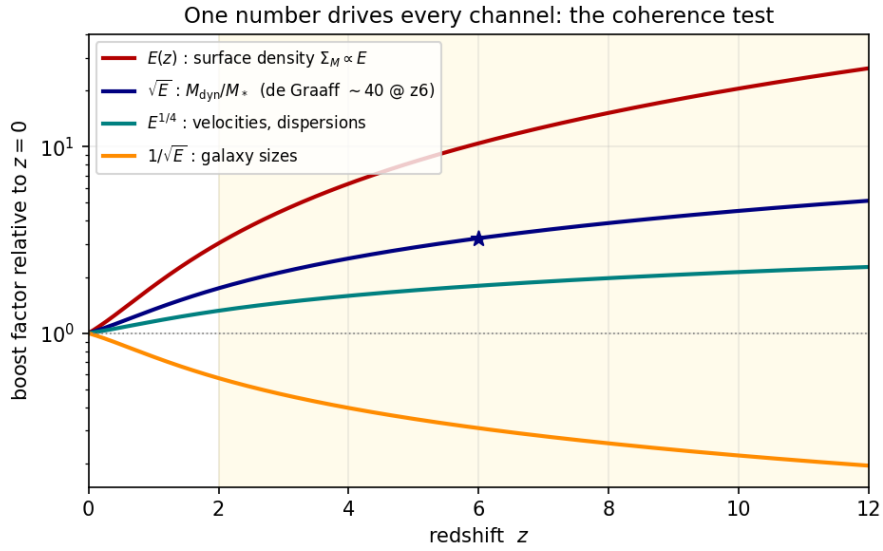


Figure 2. Each JWST observable is a different power of the SAME quantity $E(z)$, so the whole prediction set rises and falls together with one number. The star marks de Graaff (2024) $M_{\text{dyn}}/M_* \approx 40$ at $z \approx 6$. Coherence across channels is the signature; independent scatter would refute it. [jwst_full_predictions.py]

The predictions

Dynamical masses rise as $\sqrt{E(z)}$. In MOND the apparent “dark matter” of a galaxy — the gap between its kinematic mass and its baryonic mass — grows with a_0 . So the ratio M_{dyn}/M_* measured from JWST spectroscopy should climb with redshift as $\sqrt{E}$: a factor of three by $z \approx 6$, four and a half by $z \approx 10$. This is already hinted at: de Graaff and collaborators (2024) find dynamical-to-stellar ratios as high as 40 in JADES galaxies at $z \approx 5.5\text{--}7.4$, values that an unchanging a_0 struggles to reach.

The Tully-Fisher and Faber-Jackson relations shift. Because $v^4 = G \cdot M \cdot a_0(z)$, a galaxy of fixed baryonic mass spins faster at high redshift, and the zero-point of the mass-velocity relation moves by $-\log_{10} E(z)$ — roughly half a dex by $z = 2$, a full dex by $z = 6$. The intermediate-redshift precedent already exists: Übler and collaborators (2017) measured a shift of −0.45 dex at $z \approx 2.3$, almost exactly the −0.48 the relation predicts.

Velocities and dispersions are boosted as $E(z)^{1/4}$, about +80% by $z = 6$ at fixed baryonic mass. **The radial-acceleration relation's knee moves to higher acceleration.** Galaxies enter

the deep-MOND regime at **smaller radius** (the characteristic scale shrinks as $1/\sqrt{E}$), and their **critical surface density rises as $E(z)$** , so early disks should be denser.

One clean null test. Build a galaxy scaling relation from a sample spanning a range of redshifts. If a_0 evolves, the relation is artificially *broadened* — by an amount $\log E(z_{\max})$ — because each galaxy carries a different a_0 . Rescale each by its own $E(z)$ and the relation should snap back to its intrinsic, tight form. Do-nothing-broadens, rescaling-tightens: a built-in falsifier that needs no external calibration.

What this does not predict — and why that matters

A theory is only as trustworthy as the claims it refuses to make. The evolving a_0 is **absent from the linear growth of cosmic structure** (this follows from the structure of the only relativistic completion that fits the microwave background). The consequence is important and deflationary: **evolving a_0 does not, by itself, make galaxies form earlier or in greater number.** It changes the *dynamics* of galaxies that have already formed, not the *census* of how many exist. So the much-discussed “impossible” early massive galaxies are addressed by this framework **only** where their masses are inferred dynamically — in which case those masses are over-estimated by $\sqrt{E(z)}$, and the galaxies are lighter in baryons than they appear. Where a mass excess comes from stellar light, or where the puzzle is the sheer abundance of luminous sources, this framework is **silent**. It likewise says nothing about metallicities, reionization, dust, or the ultraviolet luminosity function — those are the physics of gas and stars, not of a_0 . Naming them as victories, as earlier versions of this work did, was a mistake. The honest boundary is part of the prediction.

How to run the test

Do not fit a single galaxy. **Measure the dynamical-mass ratio, the Tully-Fisher zero-point, the dispersions, the sizes, and the surface densities across a span of redshift, and ask whether they all key off the same $E(z)$.** If $M_{\text{dyn}}/M_* \propto \sqrt{E}$ and the zero-point $\propto -\log E$ and the dispersions $\propto E^{1/4}$, simultaneously, with one consistent expansion history — that coherence is the fingerprint of an evolving a_0 , and nothing else makes it. If the channels scatter independently, the idea is wrong, and the same data say so cleanly. The single most decisive measurement is a clean determination of $M_{\text{dyn}}/M_{\text{baryon}}$ at a known baryonic acceleration in a galaxy at $z > 2$.

What is actually being claimed

Not a theory of everything; not a derivation of the MOND scale; not a solution to the dozens of problems earlier drafts of this idea claimed. The claim is narrow and, for that reason, testable: **that the acceleration scale governing galaxies is the cosmic density's own acceleration, so it must evolve as $E(z)$ — and JWST is now precise enough to see it.** The universe's largest scale would then be reaching into its smallest dynamics: *here* would be set by *everywhere*. That is either true or it isn't, and the next few years of JWST kinematics will decide it.

Reproducibility: `a0_decisive_pipeline.py` (the 5σ data fit), `jwst_full_predictions.py` (the full prediction map), `bridge1_aest_equations.md` (why a_0 is absent from linear growth), and the repository at github.com/carlzimmerman/zimmerman-formula. *Key data:* McGaugh, Lelli & Schombert 2016 (SPARC); Übler et al. 2017 (KMOS^{3D}); de Graaff et al. 2024 (JADES); Vărășteanu et al. 2025; MUSE-DARK III 2026. *Foundations:* Milgrom 1983; Skordis & Złośnik 2021 (the relativistic completion); Verlinde 2017.